

On Large Deviations in the ABC Conjecture and Prime Factor Constraints

Thomas Hoffbauer

Abstract

We study large deviations in the ABC conjecture via the quantity $\Delta(a, b, c) := \log c - \log \text{rad}(abc)$. Numerical experiments indicate a strong negative correlation between Δ and the number $\omega(abc)$ of distinct prime factors.

We formulate a structural hypothesis stating that large values of Δ can only occur when the prime support is small, and provide a partial theoretical explanation based on S-unit equations and bounds from Baker theory.

1 Introduction

The ABC conjecture states that for coprime integers a, b with $a + b = c$, and for every $\varepsilon > 0$, there exists K_ε such that $c \leq K_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon}$.

We define $\Delta(a, b, c) := \log c - \log \text{rad}(abc)$.

The conjecture implies that Δ cannot grow significantly.

2 Numerical Evidence

Computations for coprime pairs (a, b) with $a, b \leq N$ (up to several million) show:

- Δ is typically negative,
- large values of Δ are rare,
- large Δ occurs only when $\omega(abc)$ is small.

2.1 Structural observation

In all observed cases with large Δ :

- the number of distinct prime factors is small,
- high prime powers occur,
- c is often close to a prime power.

These observations motivate the following hypothesis.

3 Structural Constraint

Lemma 1 (Structural Constraint). *For every $\delta > 0$, there exists a constant $K(\delta)$ such that $[\Delta(a, b, c) \geq \delta; \Rightarrow; \omega(abc) \leq K(\delta)].$*

Remark 1. *This lemma is not proven here. It is supported by numerical evidence and consistency with known diophantine constraints.*

4 Connection to S-Unit Equations

Let $[S := p : p \mid abc, \quad s = |S|.]$

Define $[x = a \frac{y}{c}, \quad y = \frac{b}{c}.] \text{ Then } [x+y=1,] \text{ where } x, y \text{ are } S - \text{units}.$

By classical results (see [2]), the number of such solutions is finite for fixed S .

5 Exponent Constraints from Baker Theory

We write $[c = \prod_{p \in S} p^{\gamma_p}.]$

Bounds from Baker theory imply that the exponents γ_p cannot grow arbitrarily when the size $s = |S|$ increases.

More precisely, there exists a function $E(s)$ with at most exponential growth such that $[\max_{p \in S} \gamma_p \leq E(s).]$

Proposition 1 (Heuristic exponent bound). *For solutions of $a + b = c$ with prime support S , one has $[\Delta(a, b, c) \leq (E(s) - 1) \sum_{p \in S} \log p.]$*

Heuristic argument. We write $[\Delta = \sum_{p \in S} (\gamma_p - 1) \log p.]$ If s is large, Baker-type bounds imply that most exponents γ_p must remain small, preventing large contributions to Δ . $\square$

6 Heuristic Consequences

Using the prime number theorem, $[\sum_{p \in S} \log p \gtrsim s \log s.]$

Thus, $[\Delta \lesssim E(s), s \log s.]$

Since $E(s)$ grows at most exponentially, this suggests that large Δ can only occur when s is small.

7 Quantitative Behaviour

The above argument suggests the existence of a monotone function $K(\delta)$ such that $[\Delta \geq \delta; \Rightarrow; \omega(abc) \leq K(\delta),]$ with $K(\delta)$ growing slowly as $\delta \rightarrow 0$.

8 Discussion

This perspective suggests that large deviations in the ABC conjecture correspond to highly constrained configurations of prime factors.

In particular, large Δ appears to require strong concentration of multiplicative structure in a small number of primes.

9 Outlook

A possible route toward the ABC conjecture is to rigorously establish the structural constraint or to derive it from refined diophantine bounds.

References

- [1] A. Baker and G. Wüstholz, *Logarithmic Forms and Diophantine Geometry*, Cambridge University Press, 2007.
- [2] J.-H. Evertse and K. Győry, *Unit Equations in Diophantine Number Theory*, Cambridge University Press, 2015.